GOURAV SINGH

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PROFESSIONAL SUMMARY

Early-career Software Engineer pursuing a B.E. in Robotics and Artificial Intelligence Engineering with a strong foundation in Computer Science and Mathematics. Demonstrated hands-on experience through multi-person projects, including a scalable full-stack e-commerce platform and an AI-powered drug discovery pipeline. Proficient in **JavaScript** and **React**, focusing on hands-on experience in hackathon-winning projects.

EDUCATIONS

Bangalore Institute of Technology | Bengaluru — 2023 - 2027
B.E. in Robotics & Artificial Intelligence — *CGPA : 8.55/10*

Kendriya Vidalaya No.2 BSF | Indore — 2020 - 2022
CBSE Class XII,Class X — *GRADES : XII-92.6%,X-92.6%*

PROJECTS

Argus — Edge-Native Working-Capital Decision Engine |

- Designed and built **Argus**, a working-capital decision engine, utilizing **React** and **TypeScript**, deployed serverlessly on **Cloudflare Workers** with **Cloudflare D1** to store compact, immutable analysis snapshots and record user-approved outcomes.
- Developed a browser-native quantitative forecasting engine that automatically backtests **Seasonal Naive**, **Simple Exponential Smoothing**, and **Croston's Method** per SKU to select the lowest **WAPE** (Weighted Absolute Percentage Error) model, integrated with a robust forecast-residual anomaly detector using **Median Absolute Deviation** (MAD).
- Enforced financial safety by routing all cash runway, budget, and MOQ calculations through deterministic code while strictly isolating **Gemini 2.5 Flash** to explanations, secured by a numeric verification gate that discards LLM output if unsupported numbers are detected.

Technologies / Tools Used : *React, Tailwind CSS, Gemini Flash 2.5 API, Cloudflare*

ChaosLab – Chaos Engineering & Auto-Repair Platform |

- Developed a **chaos engineering platform** that executes automated fault injection, including **CPU spikes**, latency simulation, and container failures.
- Engineered an **auto-repair system** utilizing **Prometheus** metrics, featuring automated remediation workflows.
- Designed a **Node.js** control plane with **BullMQ**, **Dockerode**, **Toxiproxy**, and **React** dashboards, enabling live log streaming.

Technologies / Tools Used : *Node.js, React, Docker, Prometheus, Redis, k6*

MolGenix – AI-Powered Drug Discovery Pipeline |

- Engineered an **AI-driven drug discovery pipeline** leveraging **NLP**, **generative molecular modeling**, ADMET prediction, and molecular docking workflows.
- Developed a **FastAPI** backend, utilizing **PostgreSQL** and **Redis** for orchestration, molecule validation, and real-time processing.
- Employed **RDKit**, **DeepChem**, and **AutoDock Vina** for toxicity screening and binding affinity analysis.
- Containerized and deployed backend services with comprehensive monitoring dashboards.

Technologies / Tools Used : *React, FastAPI, PostgreSQL, Redis, TensorFlow*

SKILLS

Core Concepts :	DBMS, OOPS, System design
Databases :	PostgreSQL, Redis
Tools & Platforms :	Docker, GitHub, Postman, Prometheus
Frameworks & Libraries :	React, Node.js, Express.js
Programming Languages :	C++, Python, JavaScript
Soft Skills :	Leadership, Communication

AWARDS & ACHIEVEMENTS

- **Secured 2nd place** at HackSurgeX Hackathon for an **AI-driven drug discovery solution** at NMIT Bengaluru.
- **Won** Project Expo 2025 at BIT for **CNN-based AI project implementation**.
- **Qualified** for the SIH Hackathon with an **AI-based student dropout warning system**
- Completed over **100 LeetCode** problems across various difficulty levels.
- **Achieved** AIR 483 in the NDA 151 Course conducted by UPSC.

CERTIFICATIONS

Stanford University - Completed coursework in **Machine Learning**
Google - Earned **Generative AI Certification**
Postman - Achieved **API Fundamentals** certification
Infosys & IBM-AI Training completion certificate
TLE Eliminators - Certification in Data Structures and Algorithms

EXTRACURRICULAR ACTIVITIES

- **Active member** of **Aero Club**, participating in **RC aircraft construction workshops** and leading outreach initiatives.
- **Competed** in national-level sports competitions.
- Engaged in **public speaking**, **debates**, and **quizzes**.